

Final Model Recommendation

Model choice depends on the validation objective

Best primary IR model

E11

YOLO11l-seg

EO+IR training, image size 960

IR mask mAP50-95

0.5250

Best combined EO+IR model

E12

YOLO11x-seg

EO+IR training, image size 960

EO+IR mask mAP50-95

0.4422

E11 is preferred for strict IR segmentation; E12 is preferred for combined-modality robustness.